

José R. Ocampo-Gómez

Curriculum Vitae

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RESEARCH EXPERIENCE

JUNE 2025 - PRESENT

Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)

Undergraduate Researcher

AGN Reverberation Mapping:

- Advisor: Prof. Steven Allen

- Measured reverberation time lags between AGN continuum and Fe II broad-line emission across 25 OzDES-monitored active galactic nuclei ($z \approx 1 - 2$), to constrain the spatial extent of the Fe II emission region as a probe of supermassive black hole properties.

- Applied and cross-validated three independent time-series techniques (PyCCF, JAVELIN, & PyROA) on multi-year, irregularly-sampled photometric and spectroscopic data.

- Developed a simulation-based validation framework to quantify each method's accuracy and failure modes under realistic noise and cadence.

- Developing constraints on the Fe II radius-luminosity relation to estimate black hole masses.

AGN Luminosity Evolution:

- Advisor: Prof. Vahé Petrosian

- Determining the cosmological evolution of the gamma-ray and radio luminosity functions of flat spectrum radio quasars (FSRQs) using non-parametric methods for truncated data.

- Constructing a tri-variate sample from Fermi-LAT gamma-ray, VLBI radio, and Gaia DR2 optical observations, accounting for the additional selection bias introduced by optical redshift measurement.

- Applying the Efron-Petrosian method and Lynden-Bell C^- estimator to recover unbiased luminosity evolution indices and local luminosity functions.

- Measuring the intrinsic gamma-ray-radio luminosity correlation to constrain the relationship between jet and accretion-disk emission.

APRIL 2023 - APRIL 2024

NASA California Space Grant Consortium

Undergraduate Researcher

- Advisor: Prof. Jayesh Bhakta

- Built a low-cost flight telemetry and data-logging system for experimental RC aircraft, integrating inertial, barometric, GPS, and current sensors on a Raspberry Pi Pico in MicroPython.

- Implemented a packet-based nRF24Lo1 downlink with redundant on-board microSD logging to preserve data through radio-link loss.

- Developed a real-time ground-station display in Java/Processing, deriving aircraft pitch and roll from three-axis accelerometer data.

- Identified RF interference between the aircraft's control and telemetry transmissions as the source of downlink instability.

EDUCATION

EXP. 2027 Bachelor of Science

GPA: 3.52

Physics (Core Pathway)

Stanford University

2020 - 2024 Associate of Science for Transfer

SUMMA CUM LAUDE (GPA: 4.00)

Physics, Mathematics, Engineering, & Philosophy

Los Angeles City College

PUBLICATIONS & MANUSCRIPTS

Ocampo-Gómez, J. R., & Petrosian, V. (2026). Cosmological evolution of flat-spectrum radio quasars and the intrinsic gamma-ray-radio luminosity correlation. *In preparation for ApJ.*

Ocampo-Gómez, J. R., Yu, Z., & Allen, S. W. (2026). Fe II reverberation lags in OzDES active galactic nuclei. *In preparation.*

FELLOWSHIPS & AWARDS

2026 VPUE STEM Fellow

Stanford University

2025 Physics SURP Funding

Stanford University

2024 Jack Kent Cooke Transfer Scholarship

Jack Kent Cooke Foundation

2023 Physics & Engineering Department Scholarship

Los Angeles City College

2023 Mathematics Department Scholarship

Los Angeles City College

2022 Eugene McKnight Philosophy Scholarship

Los Angeles City College

POSTER PRESENTATIONS

Stanford Physics Summer Undergraduate Research Program (SURP) (2026). The cosmological evolution of gamma-ray and radio luminosity functions in FSRQs

Stanford Symposium of Undergraduate Research and Public Service (SURPS) (2026); Stanford Physics SURP (2025). Probing the Fe II emission region of AGNs with the OzDES RM project

National Conference for Undergraduate Research (NCUR) (2024); Southern California Conference for Undergraduate Research (SCCUR) (2023). Flight data telemetry device with data logging using customized off-the-shelf modules for experimental radio control aircraft

OUTREACH & SERVICE

SEPT. 2025 - PRESENT

KIPAC Public Outreach *Team Member*

- Presented career-focused talks and led scientific demonstrations for underrepresented high school students
- Facilitated public stargazing nights at the Stanford Student Observatory, operating telescopes, managing attendee flow, and engaging visitors in discussions of astronomy
- Delivered astronomy talks in Spanish as part of the Noches Astronómicas outreach program, including moderated Q&A sessions

JUNE 2026 - PRESENT

Stanford Community College Outreach Program (CCOP) *Mentor*

- Mentored community college students on the STEM transfer process, attending program workshops alongside them and leading follow-up discussions in small groups of eight to ten.
- Advised mentees on applications, coursework, and program selection, and connected them with resources specific to transferring into STEM fields.

APRIL 2023 - JULY 2023

Go Baby Go, Los Angeles City College *Technician (Volunteer)*

- Built and modified electric ride-on toy cars into powered mobility devices for six toddlers (ages 1.5-4) with disabilities, primarily Down syndrome, in collaboration with West Coast University physical therapy graduate students.
- Fitted each car to the individual child based on a parent questionnaire, with modifications ranging from rewiring the foot pedal to a hand-operated button and adding parent remote control to reduced top speed and added trunk, shoulder, and head support.

LEADERSHIP POSITIONS

MARCH 2026 - PRESENT

Stanford SSI *Policy Team Co-Lead*

- Leading a project to draft a model regulatory framework for amateur and high-power rocketry in Uruguay, an activity currently outside any dedicated national licensing or airspace regime.
- Assessing the case for Uruguay as a regional hub for amateur rocketry, addressing launch-site siting, third-party liability, and state responsibility obligations under Article VI of the Outer Space Treaty.
- Co-authored a policy paper on preventing the deployment of nuclear weapons in orbit; leading additional work on satellite traffic management.
- Managing small research teams and setting the agenda across international space law, space security, and nuclear nonproliferation.

JUNE 2026 - PRESENT

Stanford Transfer Network (STN) *Co-President*

- Helped promote community among transfer students and alumni.
- In charge of finances and budgeting for social events, as well as communication and planning.

TALKS

Invited Research Talk - MESA Program, College of San Mateo Planetarium (2026). Listening to the Echoes of Supermassive Black Holes.

TRiO CenCal Conference, Evergreen Valley College - KIPAC Public Outreach (2026). Magic with Lights.

Noches Astronómicas - KIPAC Public Outreach (2025). Agujeros Negros: Los motores invisibles del Universo.

PROJECTS

Bayesian AGN SED Modeling (Stanford Physics 113, 2026)

- Developed a Python implementation of thin-disk spectral energy distribution fitting for AGN.
- Applied Metropolis-Hastings MCMC to infer black hole masses and accretion rates from observational data.
- Reproduced literature measurements for high-redshift blazars and investigated sensitivity to numerical integration methods.

COMPUTER SKILLS

Python/Jupyter, C++, Mathematica, $\text{\LaTeX}$, UNIX, Git/GitHub, MicroPython, Java/Processing

COMMUNICATION SKILLS

Public Outreach, Scientific Communication, Bilingual Fluency (Spanish), Management/Supervising

STUDENT MEMBERSHIPS

American Physical Society (APS) (Since 2025)

National Society of Hispanic Physicists (NSHP) (Since 2025)

Society of Physics Students (SPS) (Since 2025)